

Second Order Critical Complements in Fabius Conditioning

Uniform overlap corrections and a strict comparison of two observation masks

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Abstract

Two recent ProvelT manuscripts study near-total observation of a geometrically weighted uniform series under a small-value constraint. They omit different bulk coordinates. We derive a uniform expansion of their overlap when the number of omitted bulk coordinates is comparable with the logarithm of the bulk size. The expansion includes the phase-dependent constant and the next inverse-complement coefficient, with a quantitative remainder. It answers the second-order crossover question in the prefix-observation manuscript and the logarithmic slice of the other manuscript's growing-complement question. The two masks have the same leading crossover but different next constants: hiding the earliest bulk coordinates gives strictly greater overlap than hiding the latest ones, eventually. An elementary signed Gamma convolution argument proves the statements without importing a uniform conditioning theorem. These are manuscript-level proofs, not a proof-assistant formalization or a certification of worldwide novelty.

1 Scope and relation to the source questions

This note examines the repository snapshot 63a7a325109ba611a1816b61dfd0eb072b896a7a. The article in *Critical Complements Sharp Information Loss* [1], Section 10, asks for the complete next-order overlap expansion when $m \sim c \log n$, including the geometric phase and integer rounding. Its Theorem 2.2 already proves the leading equivalent for every $m = o(n)$. Here [theorem 3.1](#) and [corollary 3.2](#) give a finer expansion in the critical window, including an additional $1/m$ coefficient.

The article in *Critical Complements Fabius Conditioning* [2], Section 13, instead observes a block obtained by deleting the earliest m bulk coordinates. It leaves growing m open. The block is not the prefix in [1]. We prove the same expansion for this second mask, with a different analytic factor, and establish the strict comparison in [corollary 3.3](#). We do not claim that the leading theorem for one mask previously solved the question for the other.

The probability representation of the Fabius function is classical [3]. Exponential tilting, Gamma asymptotics, and precise-deviation methods are also established tools; compare [4, 5]. The proposed contribution here is the explicit clipped-density expansion and its application to these two specified masks. The literature check was targeted, not exhaustive. No famous conjecture, publication priority, or Lean verification is claimed.

2 The geometric model and the two masks

Fix $q \in (0, 1)$ and $\rho > 0$. Let

$$S_q = \sum_{j \geq 1} q^j U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1], \quad t_n = \rho q^{-(n+1)}. \quad (2.1)$$

Write P for the product law, $T_n = t_n S_q$, and define

$$\frac{dQ_n}{dP} = \frac{e^{-T_n}}{\mathbb{E}_P e^{-T_n}}, \quad \mu_n = \mathbb{E}_{Q_n} T_n, \quad P_n = P(\cdot \mid T_n \leq \mu_n). \quad (2.2)$$

The constraint is an inequality under the original law. Under Q_n , the scaled coordinates are independent with truncated exponential density

$$p_a(x) = \frac{e^{-x}}{1 - e^{-a}} \mathbf{1}_{(0,a)}(x), \quad a = \rho q^{j-(n+1)}. \quad (2.3)$$

The first n coordinates form the bulk. The boundary tail starts at coordinate $n+1$, of cap ρ . Set

$$a_r = \rho q^{-r} \quad (r \geq 1), \quad R_0 = \sum_{\ell \geq 0} Y_\ell, \quad Y_\ell \sim p_{\rho q^\ell} \quad \text{independently.} \quad (2.4)$$

Thus $0 \leq R_0 \leq \rho/(1-q)$. For $1 \leq m < n$, define two observed sets:

$$B_{n,m}^{\text{late}} = \{1, \dots, n-m\}, \quad B_{n,m}^{\text{early}} = \{m+1, \dots, n\}. \quad (2.5)$$

The superscript names the hidden bulk coordinates. Both masks hide the whole infinite boundary tail.

If $P_{n,m}^j, Q_{n,m}^j$ are the corresponding vector marginal laws, let

$$\mathcal{O}_{n,m}^j = 1 - d_{\text{TV}}(P_{n,m}^j, Q_{n,m}^j), \quad \varepsilon_n = (2\pi n)^{-1/2}, \quad j \in \{\text{early}, \text{late}\}. \quad (2.6)$$

Total variation is $\sup_A |P(A) - Q(A)|$, so overlap is $\int \min(dP, dQ)$.

The phase factors needed below are

$$H_0(s) = \mathbb{E} e^{sR_0}, \quad (2.7)$$

$$\mathcal{H}_{\text{early}}(s) = H_0(s), \quad (2.8)$$

$$\mathcal{H}_{\text{late}}(s) = H_0(s) \prod_{r \geq 1} \frac{1 - e^{-(1-s)a_r}}{1 - e^{-a_r}}, \quad s < 1. \quad (2.9)$$

The product converges locally uniformly, with every derivative, in the complex half-plane $\Re s < 1$. Both factors are strictly positive for real $s < 1$ and equal one at zero. If desired, H_0 itself has the explicit product

$$H_0(s) = \prod_{\ell \geq 0} \frac{1 - e^{-(1-s)\rho q^\ell}}{(1-s)(1 - e^{-\rho q^\ell})}, \quad (2.10)$$

with each removable value interpreted continuously.

The first source [2] uses $t_n = \rho_{\text{first}} q^{-n}$. The exact phase match to our convention is $\rho_{\text{first}} = \rho/q$. No asymptotic reindexing is hidden in this comparison.

3 The uniform expansion

Let

$$I(y) = y - 1 - \log y, \quad s(y) = I'(y) = 1 - \frac{1}{y}, \quad d = \frac{1}{2} \log \frac{n}{m}, \quad z = \frac{d}{m}. \quad (3.1)$$

The two roots of $I(y) = z$ are denoted $y_- < 1 < y_+$. For a positive analytic factor $\mathcal{H}$ from (2.8) or (2.9), define

$$L(y) = \log \mathcal{H}(s(y)), \quad (3.2)$$

$$A(y) = -\frac{1}{12} - \frac{\mathcal{H}''(s(y))}{2y^2 \mathcal{H}(s(y))}, \quad (3.3)$$

$$F(y) = \frac{L(y) + 1}{s(y)}, \quad (3.4)$$

$$B(y) = \frac{A(y)}{s(y)} + \frac{(L(y) + 1)L'(y)}{s(y)^2} - \frac{L(y)^2/2 + L(y) + 1}{y^2 s(y)^3}. \quad (3.5)$$

Here $\mathcal{H}''$ is differentiated in s and L' in y ; in particular $L'(y) = \mathcal{H}'(s(y))/(y^2 \mathcal{H}(s(y)))$. We write $[G]_{y_-}^{y_+} = G(y_+) - G(y_-)$.

Theorem 3.1 (Critical crossover through the inverse-complement term). *Fix q, ρ and $0 < c_0 < c_1 < \infty$. Uniformly over integers m such that $c_0 \log n \leq m \leq c_1 \log n$, as $n \rightarrow \infty$, one has*

$$\frac{\mathcal{O}_{n,m}^j}{\varepsilon_n} = m(y_+ - y_-) + [F_j]_{y_-}^{y_+} + \frac{1}{m}[B_j]_{y_-}^{y_+} + O\left(\frac{1}{m^2} + \frac{m^3}{n}\right). \quad (3.6)$$

The factors in (2.8)–(2.9) determine F_j, B_j . All roots use the actual integer m and $d = \frac{1}{2} \log(n/m)$. Consequently the same formula with the $1/m$ term omitted has remainder $O(1/m)$.

The geometric phase first appears in the bounded correction $[F_j]$; the leading term forgets it. The two exterior tails of the clipped density each contribute a nonzero part of this correction. Keeping only the distance between likelihood crossings would omit $[1/s]_{y_-}^{y_+}$.

3.1 The logarithmic and rounding terms

For $c > 0$, let $Y_{\pm}(c)$ solve $I(Y) = 1/(2c)$ and set

$$W(c) = Y_+(c) - Y_-(c), \quad C(c) = cW(c), \quad (3.7)$$

$$D(c) = \frac{1}{s(Y_+(c))} - \frac{1}{s(Y_-(c))}, \quad K_j(c) = [F_j]_{Y_-(c)}^{Y_+(c)}. \quad (3.8)$$

Corollary 3.2 (Explicit logarithmic correction). *Put $\ell = \log n$ and $c_n = m/\ell$. Uniformly for $c_n \in [c_0, c_1]$,*

$$\frac{\mathcal{O}_{n,m}^j}{\varepsilon_n} = \ell C(c_n) - \frac{1}{2} D(c_n) \log \ell + K_j(c_n) - \frac{1}{2} D(c_n) \log c_n + O\left(\frac{(\log \ell)^2}{\ell} + \frac{\ell^3}{n}\right). \quad (3.9)$$

If $m = c\ell + \beta_n$ for fixed $c > 0$ and bounded β_n , then

$$\begin{aligned} \frac{\mathcal{O}_{n,m}^j}{\varepsilon_n} &= \ell C(c) - \frac{1}{2} D(c) \log \ell + \beta_n \left(W(c) - \frac{D(c)}{2c} \right) \\ &\quad + K_j(c) - \frac{1}{2} D(c) \log c + o(1). \end{aligned} \quad (3.10)$$

Thus rounding $m = \lfloor c \log n \rfloor$ generally produces an oscillating bounded correction. The assumption $m \sim c \log n$ alone does not permit replacing c_n by c in (3.9) to constant accuracy.

Proof. If $\Delta(z) = y_+(z) - y_-(z)$, implicit differentiation gives $\Delta'(z) = 1/s(y_+) - 1/s(y_-)$. Since $z = 1/(2c_n) - \log(c_n \ell)/(2c_n \ell)$, Taylor's theorem applied on a compact positive interval gives

$$m\Delta(z) = \ell C(c_n) - \frac{1}{2} D(c_n) \log(c_n \ell) + O((\log \ell)^2/\ell).$$

Replacing the arguments of F_j costs $O(\log \ell/\ell)$. This proves (3.9). Finally $C'(c) = W(c) - D(c)/(2c)$, which gives (3.10). $\square$

3.2 A strict distinction between the two masks

Corollary 3.3 (Hiding earlier coordinates gives greater overlap). *Let $m/\log n \rightarrow c \in (0, \infty)$. Put*

$$\Psi(s) = \sum_{r \geq 1} \log \frac{1 - e^{-(1-s)a_r}}{1 - e^{-a_r}}. \quad (3.11)$$

Then, with $s_{\pm} = s(Y_{\pm}(c))$,

$$\frac{\mathcal{O}_{n,m}^{\text{late}} - \mathcal{O}_{n,m}^{\text{early}}}{\varepsilon_n} \longrightarrow \frac{\Psi(s_+)}{s_+} - \frac{\Psi(s_-)}{s_-} < 0. \quad (3.12)$$

In particular $\mathcal{O}_{n,m}^{\text{late}} < \mathcal{O}_{n,m}^{\text{early}}$ for all sufficiently large n along that sequence.

Proof. The leading root term and the $1/s$ tail terms cancel in the difference. Since $\log \mathcal{H}_{\text{late}} - \log \mathcal{H}_{\text{early}} = \Psi$, [theorem 3.1](#) gives the stated limit. Termwise differentiation is valid, and

$$\Psi''(s) = - \sum_{r \geq 1} \frac{a_r^2 e^{-(1-s)a_r}}{(1 - e^{-(1-s)a_r})^2} < 0, \quad \Psi(0) = 0.$$

Strict concavity implies that the secant slope on $[s_-, 0]$ is strictly larger than that on $[0, s_+]$. These slopes are $\Psi(s_-)/s_-$ and $\Psi(s_+)/s_+$, respectively. $\square$

4 A clipped signed Gamma transfer theorem

The following analytic statement isolates the calculation. It is deliberately stated with hypotheses that can be checked directly for both masks. Write

$$\gamma_k(x) = \frac{e^{-x} x^{k-1}}{\Gamma(k)} \mathbf{1}_{(0,\infty)}(x), \quad \|\eta\|_{\theta} = \int_0^{\infty} e^{\theta u} |\eta|(du).$$

Lemma 4.1 (Density and endpoint tail expansions). *Let η_m be real finite signed measures supported in $[0, \infty)$, of mass one. Suppose, for every $\theta < 1$,*

$$\sup_m \|\eta_m\|_{\theta} < \infty, \quad \|\eta_m - \eta\|_{\theta} = O(m^{-2}),$$

*and put $\mathcal{H}(s) = \int e^{su} \eta(du)$. Suppose $\mathcal{H}(s) > 0$ for real $s < 1$ and $g_m = \gamma_{m+1} * \eta_m$ is a nonnegative probability density. Uniformly for y in compact subsets of $(0, \infty)$,*

$$g_m(my) = \frac{e^{-mI(y)\mathcal{H}(s(y))}}{\sqrt{2\pi m}} \left(1 + \frac{A(y)}{m} + O(m^{-2}) \right). \quad (4.1)$$

Here A is (3.3). If the convergence of η_m above is faster than every inverse power, the hypothesis is in particular satisfied.

Proof. The exact Gamma ratio is

$$\frac{\gamma_{m+1}(my - u)}{\gamma_{m+1}(my)} = e^u \left(1 - \frac{u}{my} \right)^m \mathbf{1}_{\{u < my\}}. \quad (4.2)$$

For y in a fixed compact interval and $0 \leq u \leq my/2$, expanding $\log(1 - u/(my))$ and then the exponential gives

$$e^{s(y)u} \left(1 - \frac{u^2}{2my^2} \right) + O\left(\frac{(u^3 + u^4)e^{s(y)u}}{m^2} \right). \quad (4.3)$$

For completeness, the extra negative exponent beyond its quadratic term is bounded by Cu^3/m^2 on this interval. The inequalities $|e^{-v} - 1 + v| \leq v^2/2$ and $1 - e^{-w} \leq w$ for $v, w \geq 0$ prove (4.3) with an absolute bound of the displayed form. On $u > my/2$, both the exact ratio and the approximating terms are exponentially negligible after integration: choose $\theta \in (\max_y s(y), 1)$ and use the weighted variation bound. Thus integration against the signed measure gives

$$\int \frac{\gamma_{m+1}(my - u)}{\gamma_{m+1}(my)} \eta_m(du) = \mathcal{H}(s(y)) - \frac{\mathcal{H}''(s(y))}{2my^2} + O(m^{-2}).$$

The $O(m^{-2})$ convergence hypothesis permits replacement of all needed weighted moments by those of η ; polynomial factors are absorbed using a slightly larger exponent $\theta < 1$. Stirling's formula gives

$$\gamma_{m+1}(my) = \frac{e^{-mI(y)}}{\sqrt{2\pi m}} \left(1 - \frac{1}{12m} + O(m^{-2}) \right).$$

Multiplication proves (4.1). Positivity of $\mathcal{H}$ on compacts justifies the relative error and its logarithm. $\square$

Theorem 4.2 (Clipped profile expansion). *In addition to lemma 4.1, suppose each g_m is continuous and log-concave on $(0, \infty)$, tends to zero at both endpoints, and has positive interior values. For z in a fixed compact subset of $(0, \infty)$, put*

$$a = (2\pi m)^{-1/2} e^{-mz}, \quad J_m(a) = \int_0^\infty \min\{1, g_m(u)/a\} du.$$

Then, uniformly in that set of z ,

$$J_m(a) = m(y_+ - y_-) + [F]_{y_-}^{y_+} + \frac{1}{m} [B]_{y_-}^{y_+} + O(m^{-2}). \quad (4.4)$$

Proof. All roots remain in compact subsets of $(0, 1)$ and $(1, \infty)$. Consequently $s(y_\pm)$ are uniformly separated from zero. Log-concavity makes the superlevel set $\{g_m \geq a\}$ an interval; lemma 4.1 places its two endpoints near $my_\pm$. A nonconstant log-concave density cannot have a flat segment at a level below its maximum: a flat segment would be a set of maximizers. Thus its endpoints are the unique crossings, denoted $u_\pm$.

Taking logarithms in (4.1) gives

$$\log(g_m(my)/a) = m(z - I(y)) + L(y) + A(y)/m + O(m^{-2}).$$

First this implies $u_\pm = my_\pm + O(1)$. Successive Taylor expansion then gives

$$u_\pm = my_\pm + b(y_\pm) + \frac{c(y_\pm)}{m} + O(m^{-2}), \quad (4.5)$$

$$b(y) = \frac{L(y)}{s(y)}, \quad c(y) = \frac{A(y) + L'(y)b(y) - b(y)^2/(2y^2)}{s(y)}. \quad (4.6)$$

These conclusions need no derivative estimate on the saddle remainder: the uniform $O(m^{-2})$ error and the nonzero slope bracket each crossing within $O(m^{-2})$ of its candidate position.

We next retain both exterior tails. If v stays in a compact subset of $(1, \infty)$, integration by parts in the saddle formula gives

$$\frac{\int_{mv}^\infty g_m(u) du}{g_m(mv)} = \frac{1}{s(v)} + \frac{1}{m} \left(\frac{L'(v)}{s(v)^2} - \frac{1}{v^2 s(v)^3} \right) + O(m^{-2}). \quad (4.7)$$

On a compact subset of $(0, 1)$ the corresponding lower-tail formula is the negative of the right-hand expression:

$$\frac{\int_0^{mv} g_m(u) du}{g_m(mv)} = -\frac{1}{s(v)} - \frac{1}{m} \left(\frac{L'(v)}{s(v)^2} - \frac{1}{v^2 s(v)^3} \right) + O(m^{-2}). \quad (4.8)$$

Here is a justification including the remote tails. The moment-generating function is $(1 - s)^{-(m+1)} \int e^{su} \eta_m(du)$ for real $s < 1$. Markov's inequality, applied to the actual nonnegative law g_m , bounds the upper tail beyond mM by $C_M e^{-mI(M)}$, and the lower tail below $m\delta$ by $C_\delta e^{-mI(\delta)}$. Choose M large and δ small so that these exponents exceed the largest relevant endpoint exponent by a fixed positive amount. Both remote tails, divided by the endpoint density, are exponentially

negligible. On the remaining compact intervals, (4.1) has relative remainder $O(m^{-2})$, whose integral is $O(m^{-2})$ times the leading tail integral. For its smooth main part, integrate by parts twice using $I' = s$, $I'' = 1/y^2$. The endpoint terms are $1/s$ and $(L'/s^2 - I''/s^3)/m$. A third integration, or an absolute bound on the remaining derivative integral, gives the uniform $O(m^{-2})$ remainder. The amplitude A/m cancels with its endpoint normalization to this order. This proves both tail formulas, with the lower endpoint orientation supplying the minus sign in (4.8).

At $v = u_{\pm}/m$, substitute (4.5) and use $g_m(u_{\pm}) = a$. In a signed endpoint notation, the tail contribution is

$$\left[\frac{1}{s(y)} + \frac{1}{m} \left\{ \frac{L'(y) - b(y)/y^2}{s(y)^2} - \frac{1}{y^2 s(y)^3} \right\} \right]_{y_-}^{y_+} + O(m^{-2}).$$

Adding the interval length $u_+ - u_-$ gives (4.4), because $b + 1/s = F$ and

$$c + \frac{L' - b/y^2}{s^2} - \frac{1}{y^2 s^3} = \frac{A}{s} + \frac{(L+1)L'}{s^2} - \frac{L^2/2 + L + 1}{y^2 s^3} = B.$$

□

5 Verification for the geometric series

5.1 Exact signed convolution and its bounds

For $a > 0$ set

$$\nu_a = \frac{\delta_0 - e^{-a}\delta_a}{1 - e^{-a}}. \quad (5.1)$$

The identity $p_a = \gamma_1 * \nu_a$ follows by direct substitution. The signed measure is an analytic device, never the law of a random coordinate. For any subset of indices and any fixed $\theta < 1$,

$$\left\| \prod_{r \in D}^* \nu_{a_r} \right\|_{\theta} \leq \prod_{r \geq 1} \frac{1 + e^{-(1-\theta)a_r}}{1 - e^{-a_r}} < \infty. \quad (5.2)$$

The same bound holds for $\theta < 0$, using support in $[0, \infty)$. Since a_r grows geometrically, deleting the factors with $r > N$ changes the product by $O(\sum_{r>N} e^{-(1-\theta)a_r})$ in this norm. This error is smaller than every inverse power of N .

Let $C_{n,m}^j$ be the sum of the hidden coordinates under Q_n and let E be an independent rate-one exponential. Its convolution density $g_{n,m}^j$, the density of $C_{n,m}^j + E$, satisfies

$$g_{n,m}^{\text{late}} = \gamma_{m+1} * \eta_m^{\text{late}}, \quad \eta_m^{\text{late}} = \mathcal{L}(R_0) * \bigstar_{r=1}^m \nu_{a_r}, \quad (5.3)$$

$$g_{n,m}^{\text{early}} = \gamma_{m+1} * \eta_{n,m}^{\text{early}}, \quad \eta_{n,m}^{\text{early}} = \mathcal{L}(R_0) * \bigstar_{r=n-m+1}^n \nu_{a_r}. \quad (5.4)$$

For the late mask the limiting signed transform is (2.9). For the early mask it is (2.8); the convergence error is superexponentially small in $n - m$. Both satisfy the weighted hypotheses of lemma 4.1, uniformly in the critical window. The transfer theorem permits such a two-index family whenever its bounds are uniform.

Each p_a and the exponential density is log-concave. Finite convolution preserves log-concavity [6]. Infinite convolution here follows by weak limits; alternatively, convolve a finite partial sum with E , pass to the continuous limiting densities on compact positive intervals, and pass the defining log-concavity inequality to the limit. The resulting density is positive for $u > 0$, vanishes as $u \downarrow 0$ and as $u \rightarrow \infty$, and is continuous. Positivity follows by making a finite set of positive-cap coordinates small and bounding the remaining deterministic tail; continuity follows by convolution with two bounded continuous convolution factors. At the left endpoint, $g(u) \leq e^{-u} e^u P(C \leq u) \rightarrow 0$, since C has no atom at zero. Thus all hypotheses of theorem 4.2 hold.

5.2 The exact overlap identity

Let $W_{n,m}^j$ be the observed-coordinate sum, let $f_{n,m}^j$ be the density of $W_{n,m}^j - \mathbb{E}_{Q_n} W_{n,m}^j$, and let $r_{n,m}^j = \mathbb{E}_{Q_n} C_{n,m}^j$. Define the full normalizer

$$a_n = \mathbb{E}_{Q_n} [e^{T_n - \mu_n} \mathbf{1}_{\{T_n \leq \mu_n\}}]. \quad (5.5)$$

Conditioning the full likelihood from (2.2) on the observed vector gives its exact marginal likelihood

$$\frac{g_{n,m}^j(r_{n,m}^j - v)}{a_n}, \quad v = W_{n,m}^j - \mathbb{E}_{Q_n} W_{n,m}^j. \quad (5.6)$$

Indeed $e^{-u} \mathbb{E}[e^C \mathbf{1}_{\{C \leq u\}}]$ is the density of $C + E$. It follows that

$$\mathcal{O}_{n,m}^j = \int_0^\infty f_{n,m}^j(r_{n,m}^j - u) \min\{1, g_{n,m}^j(u)/a_n\} du. \quad (5.7)$$

This concerns the full vector marginal, not only an approximation of its sum.

Lemma 5.1 (Uniform flattening in the critical window). *For $c_0 \log n \leq m \leq c_1 \log n$ and either mask,*

$$\frac{\mathcal{O}_{n,m}^j}{\varepsilon_n} = \int_0^\infty \min\{1, g_{n,m}^j(u)/\varepsilon_n\} du + O(m^3/n). \quad (5.8)$$

Proof. The observed sum has the exact density $\gamma_{n-m} * \xi_{n,m}^j$, where $\xi^{\text{late}} = *_{r=m+1}^n \nu_{a_r}$ and $\xi^{\text{early}} = *_{r=1}^{n-m} \nu_{a_r}$. The weighted variations, all polynomial moments, and signed mean shifts are uniformly bounded by (5.2). Stirling's formula and Taylor expansion at the Gamma mean, integrated against these measures, therefore give, for every fixed M ,

$$\sup_{|v| \leq Mm} \left| \frac{f_{n,m}^j(v)}{\varepsilon_n} - 1 \right| = O(m^2/n), \quad \sup_v f_{n,m}^j(v) \leq C\varepsilon_n. \quad (5.9)$$

Here the integral can first be restricted to signed shifts $u \leq M_1 \log n$, with arbitrarily small inverse-polynomial error; on the remaining interval the elementary bound is $\sqrt{2\pi k} \gamma_k(k+w) = 1 + O((1+w^2)/k)$, where $k = n - m$ and $w = O(\log n)$. This proves the stated error for growing m , rather than invoking a fixed- m expansion.

The full normalizer is independent of the observation mask and satisfies

$$a_n = \varepsilon_n(1 + O(n^{-1})). \quad (5.10)$$

To see this directly, use the same signed representation with all n bulk variables and the fixed boundary R_0 . The centered total density, on $|v| \leq M_1 \log n$, has the local expansion $\varepsilon_n[1 + O((1+v^2)/n) + O((1+|v|^6)/n^2)]$ after integration of the bounded signed moments. Integrating this density at $v = -e$ against $e^{-e} de$, $e \geq 0$, proves (5.10); its remote tail is controlled using the global density bound. Also $r_{n,m}^j = m + O(1)$.

Choose constants $0 < \delta < 1 < M$ so that $I(\delta)$ and $I(M)$ are larger than any prescribed constant. The Chernoff estimates from the proof of [theorem 4.2](#) imply

$$P(C_{n,m}^j + E \notin [\delta m, Mm]) \leq Ce^{-Km}$$

with K arbitrarily large by this choice. Since $m \geq c_0 \log n$, the contributions outside this window, divided by ε_n , are smaller than n^{-2} if K is chosen sufficiently large. This uses the global bound in (5.9) and $a_n \asymp \varepsilon_n$. Inside the window, its length is $O(m)$ and (5.9) gives a total scaled flattening error $O(m^3/n)$.

Finally changing the clipping level from a_n to ε_n costs $O(m/n)$. In fact for $|\tau| \leq 1/2$,

$$|\min(1, x/(1+\tau)) - \min(1, x)| \leq 2|\tau| \min(1, x), \quad x \geq 0,$$

and the clipped integral is $O(m)$ by the same window and tail bounds. This proves (5.8). $\square$

Proof of theorem 3.1. Apply theorem 4.2 to (5.3) or (5.4). Since $z = \log(n/m)/(2m)$ stays in a compact subset of $(0, \infty)$, its uniformity conditions hold. Moreover $(2\pi m)^{-1/2}e^{-mz} = \varepsilon_n$ exactly. The weighted convergence errors are smaller than m^{-2} uniformly, and lemma 5.1 supplies the remaining $O(m^3/n)$ error. $\square$

6 Checks and limitations

Two exact structural checks help detect errors in the coefficients. For $\eta = \delta_b$, $b \geq 0$, translation of the Gamma density leaves J_m unchanged. Substitution of $\mathcal{H}(s) = e^{bs}$ into the formulas makes the b dependence cancel from both $[F]$ and $[B]$. For $\eta = \delta_0$, $\mathcal{H} = 1$ and

$$F = 1/s, \quad B = -\frac{1}{12s} - \frac{1}{y^2 s^3}. \quad (6.1)$$

The profile is then exactly $\text{Gamma}(m+1, 1)$, so crossings and tails can be evaluated independently with incomplete Gamma functions. The source archive includes a high-precision diagnostic for this case and the $\text{Gamma}(m+2, 1)$ case $\mathcal{H}(s) = (1-s)^{-1}$. These floating-point evaluations are checks of the formulas, not interval certificates or substitutes for the proof.

The theorem fixes q and ρ . It does not give a uniform result as $q \uparrow 1$, as $c \downarrow 0$, or as $c \uparrow \infty$. At the latter boundary the endpoint slopes approach zero and the present endpoint integration argument requires a different scaling. It does not solve the entire early-mask range $m = o(n)$, arbitrary masks, general weights, or the endpoint transseries for a fixed complement. It concerns overlap; no new entropy theorem is asserted.

All new results here remain unformalized. No exact Lean declaration is claimed as a counterpart to theorems 3.1 and 4.2 and corollary 3.3. The required formal obligations include measure-theoretic likelihood reduction, signed convolution estimates, uniform asymptotic remainders, and level-crossing analysis. The two source articles are manuscript sources, not externally refereed results. Their exact algebra has been rederived here in the scope needed for the new statements.

7 Concrete next questions

Three bounded extensions now look natural. First, identify how the correction matches the thin-complement endpoint when $m/\log n \rightarrow 0$, where y_- approaches zero and the signed moment bounds must be used on a growing range of tilt parameters. Second, obtain a transition theorem as $m/\log n \rightarrow \infty$ by replacing the separated-endpoint expansion with a uniform central saddle analysis. Third, classify arbitrary masks using their limiting correction transforms. The strict ordering proved above suggests studying whether a general ordering of the hidden-cap profiles induces an ordering of the normalized secant correction. Each extension requires new estimates; none is implicit in the present compact-critical theorem.

References

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